Resonant angular frequency, $\omega_r = \frac{1}{\sqrt{LC}}$

$$\text{or } f_r = \frac{1}{2\pi\sqrt{LC}} \qquad (4.49)$$

At series resonance,

$$\omega_r = \frac{1}{\sqrt{LC}} \text{ or } \omega_r^2 = \frac{1}{LC}$$

$$\omega_r L = \frac{1}{\omega_r C} \text{ or}$$

$$X_L = X_C \qquad (4.50)$$

This is the condition for resonance in RLC circuit.

Since X_L and X_C are frequency dependent, the resonance condition $(X_L = X_C)$ can be achieved by varying the frequency of the applied voltage.

Effects of series resonance

When series resonance occurs, the impedance of the circuit is minimum and is equal to the resistance of the circuit. As a result of this, the current in the circuit becomes maximum. This is shown in the resonance curve drawn between current and frequency (Figure 4.49).

At resonance, the impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = R \quad \text{since} X_L = X_C$$

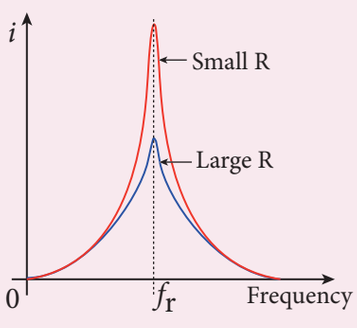


Figure 4.49 Resonance curve

Therefore, the current in the circuit is

$$I_m = \frac{V_m}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$I_m = \frac{V_m}{R} \qquad (4.51)$$

The maximum current at series resonance is limited by the resistance of the circuit. For smaller resistance, larger current with sharper curve is obtained and vice versa.

Applications of series RLC resonant circuit

RLC circuits have many applications like filter circuits, oscillators, voltage multipliers etc. An important use of series RLC resonant circuits is in the tuning circuits of radio and TV systems. The signals from many broadcasting stations at different frequencies are available in the air. To receive the signal of a particular station, tuning is done.

The tuning is commonly achieved by varying capacitance of a parallel plate variable capacitor, thereby changing the resonant frequency of the circuit. When resonant frequency is nearly equal to the frequency of the signal of the particular station, the amplitude of the current in the circuit is maximum. Thus the signal of that station alone is received.

Note

The phenomenon of electrical resonance is possible when the circuit contains both *L* and *C*. Only then the voltage across *L* and *C* cancel one another when V_L and V_C are 180° out of phase and the circuit becomes purely resistive. This implies that resonance will not occur in *RL* and *RC* circuits.

12th Physics_Vol-1_Unit 4_English Medium.indd 246 23-12-2022 17:58:39